

Susamay Garai

Senior Consultant | PwC India (PricewaterhouseCoopers Services LLP)

Mob No: +91-9088782655 | Email: susamaygarai@hotmail.com

CAREER OBJECTIVE

Aspiring cybersecurity professional with hands-on experience in penetration testing and vulnerability assessment across application, mobile (iOS and Android), web service, and thick client platforms. Seeking a challenging role in a forward-thinking organization where I can apply my technical expertise to identify and mitigate security risks, contribute to robust defense strategies, and continuously evolve in the dynamic landscape of information security. Committed to precision, ethical hacking practices, and proactive problem-solving to safeguard digital assets and enhance organizational resilience.

PROFILE SUMMARY

- ❖ Over **13.5+ years** of comprehensive IT experience spanning Dynamic Application Security Testing (DAST), Network Security, Mobile Applications (iOS and Android), Web Services, and Thick Client platforms.
- ❖ Web Applications: OWASP Top 10, business logic flaws, session management, and input validation.
- ❖ Mobile Applications (iOS & Android): Static/dynamic analysis, reverse engineering, insecure data storage, and API security.
- ❖ Web Services: SOAP, REST, authentication bypass, injection flaws, and misconfigurations
- ❖ Thick Client Applications: Memory analysis, protocol fuzzing, DLL hijacking, and binary exploitation
- ❖ Led risk assessment initiatives to maintain alignment with corporate compliance requirements.
- ❖ Conducted comprehensive security assessments on host, network, and web platforms utilizing various automated testing tools.
- ❖ Led onsite security assessments emulating insider threats to strengthen organizational defenses against internal compromise.
- ❖ Investigated hacker tactics, techniques, and procedures (TTPs), along with system-level exploits and vulnerabilities, to strengthen defensive capabilities.
- ❖ Produced detailed anomaly reports highlighting application and network security gaps, with strategic recommendations for mitigation.
- ❖ Strengthened organizational resilience by incorporating security and compliance checks, supporting continuous assurance and enhancing audit efficiency.
- ❖ Led the team to successfully deliver the project to the client within the agreed timeline.

EXPERIENCE SUMMARY

- ❖ Led penetration testing and vulnerability assessment projects as Senior Consultant at **PricewaterhouseCoopers Services LLP** from November 2019 to December 2025
- ❖ Worked as Security Test Engineer at **ARC Document Solutions Pvt. Ltd.** from August 2017 to November 2019
- ❖ Worked as Test Engineer at **STQC IT Services** under the Ministry of Electronics and Information Technology (MeitY), Government of India, from February 2012 to June 2017

TECHNICAL SKILLSET

- ❖ Web Application Security Testing with follow by the standards of OWASP Top 10
- ❖ Mobile Apps Security Testing (iOS and Android)
- ❖ Penetration Testing
- ❖ Vulnerability Assessment
- ❖ Information Security Analyst.

TOOLS USED FOR TESTING

- ❖ Web Application Security Testing Tools:
Netsparker, IBM Security AppScan, Acunetix
- ❖ Penetration Testing Tools:
Nessus Vulnerability Scanner, Sqlmap, Metasploit, Nmap (Network Mapper), Wireshark, Nikto, OpenVAS, Hping

- ❖ Mobile application Tools:
NowSecure, MobSF, Apktool, JAD, Android Virtual Device (AVD), Android Studio
- ❖ Web Proxy Tools:
Burp Suite, OWASP Zed Attack Proxy (ZAP), Paros Proxy, Webscarab
- ❖ Operating System (OS):
Kali Linux, CentOS, Red Hat, Ubuntu and Fedora, Windows

CERTIFICATION

- ❖ Certified Information Security Manager (CISM)
- ❖ Certified Information Systems Auditor (CISA)
- ❖ Certified Ethical Hacker (CEH)
- ❖ Certified Network Defender (CND)
- ❖ ISO 27001:2013 - Certified Lead Auditor

PROJECT SUMMARY

1: Project Summary: PwC Network Information Security (NIS) Project

Objective

The project aimed to identify, assess, and remediate security vulnerabilities within PwC's internal customer-facing web applications, thereby strengthening the firm's overall security posture.

Scope & Methodology

- Conducted **white-box, black-box, and gray-box penetration testing** across all internal applications.
- Applied industry-standard frameworks including **OWASP Top 10, SANS 25, NIST guidelines**, and PwC's internal security standards.
- Utilized automated scanning tools, complemented by **manual exploitation techniques** for deeper validation.

Key Findings

- Discovered vulnerabilities across **Critical, High, Medium, and Low severity levels**, including issues such as **SQL injection** and **authentication bypass**.
- Delivered **proof-of-concept exploits** and detailed **remediation steps** to facilitate effective mitigation.

Impact

- Enabled PwC's global internal applications to **patch critical flaws before exploitation**.
- Strengthened **compliance readiness** for regulatory and internal standards.
- Enhanced **customer trust** by significantly reducing the risk of data breaches.

Timeline

Each project was executed over four weeks, with the final report delivered on schedule.

2: Project Summary: ARC Document Solutions All Products

Objective

The project aimed to identify, assess, and remediate security vulnerabilities across ARC's product ecosystem—including **SKYSITE, PlanWell, ARC Abacus, Digital Storefront, and ARC Print App**—covering web applications, web services, mobile applications, and thick client platforms. The goal was to strengthen ARC's overall security posture and ensure resilience against evolving threats.

Scope & Methodology

- Conducted **white-box, black-box, and gray-box penetration testing** across all application types (Web, Mobile, API, and Thick Client).
- Applied industry-standard frameworks including **OWASP Top 10, SANS 25**, and ARC's internal security standards.
- Utilized automated scanning tools, complemented by **manual exploitation techniques** for deeper validation and accuracy.

Key Findings

- Identified vulnerabilities across **Critical, High, Medium, and Low severity levels**.
- Delivered **proof-of-concept exploits** and detailed **remediation guidance** to facilitate effective mitigation.

Impact

- Enabled ARC to **proactively patch critical flaws** across its product suite before exploitation.
- Enhanced **customer trust and confidence** by reducing the risk of data breaches across widely used ARC products.

Timeline

Security assessments are conducted on a **continuous basis**, aligned with ARC's ongoing product development cycles, ensuring that vulnerabilities are identified and addressed in real time.

3: Project Summary: Central and State Government Applications Security

Objective

The project aimed to identify, assess, and remediate security vulnerabilities across **State and Central Government applications** hosted on the NIC server, thereby strengthening the overall cybersecurity posture of critical national systems.

Scope & Methodology

- Conducted **white-box and black-box penetration testing** across a wide range of government applications.
- Applied industry-standard frameworks including **OWASP Top 10, SANS 25, and Indian Government internal security standards**.
- Utilized automated scanning tools, complemented by **manual exploitation techniques** for deeper validation.

Key Projects Covered

The assessment encompassed a diverse portfolio of mission-critical applications, including but not limited to:

- **IRCTC (Indian Railway Catering and Tourism Corporation)** – e-Ticketing and Tourism platforms
- **UIDAI (Unique Identification Authority of India)** – Aadhaar systems
- **Oriental Bank of Commerce Internet Banking**
- **All the Cooperative bank CSB application**
- **Passport Seva Data Center & Disaster Recovery Centre**
- **POSOCO (Power System Operation Corporation Limited)**
- **NCDEX e-Markets Ltd, MSTC Mining, CoalJunction** – National trading and auction systems
- **Ministry of External Affairs** – Embassy and Consulate portals worldwide
- **Integrated Financial Management Systems (IFMS)** – Odisha, Jharkhand, West Bengal
- **Kolkata Municipal Corporation, Kolkata Traffic Police, Howrah City Police**
- **Metro Railway Kolkata, Eastern Railway, Central Railway, North Eastern Railway**
- **Ministry of Earth Sciences (IMD, Positional Astronomy Centre)**
- **Health & Family Welfare Department, Govt. of West Bengal**
- **Make in India Portal, e-District Chhattisgarh, Tripura HRMS**
- **Numerous other State and Central government portals, financial systems, and public service applications.**

Key Findings

- Identified vulnerabilities across **Critical, High, Medium, and Low severity levels**.
- Delivered **proof-of-concept exploits** and detailed **remediation steps** to facilitate effective mitigation.

Impact

- Enabled government agencies to **patch critical flaws before exploitation**, safeguarding sensitive citizen and national data.
- Strengthened **compliance readiness** with CERT-IN and internal government security standards.
- Enhanced **public trust and confidence** in digital governance platforms by reducing the risk of breaches.

Timeline

Each project was executed over **two weeks**, with final reports delivered on schedule. Continuous assessments were aligned with ongoing development cycles to ensure proactive security coverage.

Additional Note

During this tenure, I also appeared for the **CERT-IN empanelment examination** on behalf of STQC and achieved a score of **over 90%**, demonstrating strong expertise in national-level cybersecurity standards.

ACADEMIC QUALIFICATIONS

Academic	Board / University	Year of passing	Percentage/DGPA
Computer Science & Engineering (B.Tech)	W.B.U.T.[AICTE. Approved]	2011	7.48
Higher secondary	W.B.C.H.S.E.	2007	65%
Secondary	W.B.B.S.E.	2005	75%

PERSONAL DETAILS

Address	4/119 Seth Bagan Road, Dumdum, Kolkata-700030
Nationality	Indian.
